

Item / GAU — Staging & DQ Framework Analysis

Object: `Item_GAU` · Source: `raw.salesforce_item` · Staging: `staging.item_gau_latest` Framework run: **2026-07-30** (read from `dq.dq_results` / `dq.rule_execution_state` on 2026-08-01). Status: **Framework executed — final results below**. No re-run performed for this document.

1. Executive Summary

Metric	Value
Run date	2026-07-30
Raw rows (non-deleted)	30,832
Staging rows (latest per Id)	30,815
Items with duplicate raw versions	574 (latest kept)
Active items	25,833
Rules in framework	26 (GAU-001..026) — 20 active gated · 6 controlled-list report-only
PASS / FAIL (active gated)	14 PASS / 6 FAIL
Total open exceptions	1,537
Distinct affected items	1,506 (some fail > 1 rule)
Controlled-list report-only	GAU-007, GAU-008, GAU-009, GAU-015, GAU-016, GAU-017 (no assumed values — distinct values in §7b)

Severity split (open exceptions)

Severity	Count
CRITICAL	0
HIGH	965
MEDIUM	239
LOW	333
Total	1,537

2. Null Analysis (RAW vs Staging)

Reused from the EDA `Top Null Columns` (staging, 30,815 rows):

Column	Null %	Staged?	Verdict
Country__c	62.6%	Yes	concern only for active non-Pledge (955 real fails); rest expected
Status__c	12.4%	Yes	all nulls are inactive → GAU-019 passes (not a concern)
Product_Type__c	5.7%	Yes	concern for active items (GAU-014 passes → active items are covered)
5 financial totals	0– 0.01%	Yes	effectively complete

3. Evidence Snapshot

Reproducible (SSMS-ready):

```
-- Staging count + active + key nulls
SELECT (SELECT COUNT(*) FROM staging.item_gau_latest) AS staging_rows,
       SUM(CASE WHEN
LOWER(LTRIM(RTRIM(COALESCE([npsp__Active__c], 'false'))))='true' THEN 1 ELSE 0 END)
AS active_items
FROM staging.item_gau_latest;

-- Sample a staged record
SELECT TOP (2) [Id],[Name],[npsp__Active__c],[Product_Type__c],[Country__c],
[Status__c]
FROM staging.item_gau_latest ORDER BY staging_created_at DESC;

-- Framework result per rule
SELECT r.check_name, r.severity, dr.check_status, dr.failed_count
FROM dq.dq_rule_catalog r
JOIN (SELECT check_name, check_status, failed_count,
       ROW_NUMBER() OVER (PARTITION BY check_name ORDER BY checked_at DESC)
rn
      FROM dq.dq_results WHERE object_name='Item_GAU') dr
ON dr.check_name=r.check_name AND dr.rn=1
WHERE r.object_name='Item_GAU' ORDER BY r.check_name;
```

Observed: `staging_rows = 30,815`, `active_items = 25,833`.

4. Table Creation and Population

`staging.item_gau_latest` rebuilt from `raw.salesforce_item`, deduped by `Id` (latest `SystemModstamp`), excluding soft-deleted rows → **30,815 rows, 31 columns** (incl. 5 financial totals added in the latest cycle). 574 items had multiple raw versions; the latest was kept and flagged via `staging_is_duplicate`.

Deduplication logic:

```

WITH dedup AS (
  SELECT
    [Id], [IsDeleted], [SystemModstamp],
    ROW_NUMBER() OVER (
      PARTITION BY CONVERT(VARCHAR(18), [Id])
      ORDER BY COALESCE(
        TRY_CONVERT(DATETIME2(7), [SystemModstamp], 127),
        TRY_CONVERT(DATETIME2(7), [SystemModstamp])
      ) DESC
    ) AS rn
  FROM [raw].[salesforce_item]
  WHERE [Id] IS NOT NULL
)
SELECT *
FROM dedup
WHERE rn = 1
  AND COALESCE(LOWER(LTRIM(RTRIM(CONVERT(NVARCHAR(20), [IsDeleted])))), N'false')
  NOT IN (N'true', N'1', N'yes', N'y');

```

Table structure (31 columns):

#	Column	#	Column
1	row_number (dedup rank)	17	Stipulation__c
2	Id	18	Regional_Office_Code__c
3	IsDeleted	19	Total_Non_Zakat_Credit__c
4	Name	20	Total_Zakat_Credit__c
5	CurrencyIsoCode	21	Total_funds_available_sadaqa__c
6	npsp__Active__c	22	Total_funds_available_zakat__c
7	Product_Type__c	23	npsp__Total_Allocations__c
8	Programme_Category__c	24	npsp__Description__c
9	Donation_Type__c	25	SystemModstamp
10	Country__c	26	_etl_source
11	Status__c	27	_etl_source_object
12	Campaign__c	28	_etl_loaded_at_utc
13	Donation_Item_Code__c	29	staging_is_duplicate
14	Allow_Single__c	30	staging_duplicate_count
15	Allow_Recurring__c	31	staging_created_at
16	HA_Donation_Frequency__c		

5. Rule Catalog Executed (26 rules)

Rule	Severity	Check	Status	Failed
GAU-001	CRITICAL	Id not null/blank	PASS	0
GAU-002	CRITICAL	Id 15/18 chars	PASS	0
GAU-003	HIGH	Name not null/blank	PASS	0
GAU-004	MEDIUM	Active is boolean token	PASS	0
GAU-005	MEDIUM	Currency populated for active	PASS	0
GAU-006	HIGH	IsDeleted boolean token	PASS	0
GAU-007	—	Product_Type — distinct values reported (not gated)	 REPORT-ONLY	—
GAU-008	—	Programme_Category — distinct values reported (not gated)	 REPORT-ONLY	—
GAU-009	—	Donation_Type — distinct values reported (not gated)	 REPORT-ONLY	—
GAU-010	HIGH	Active non-Pledge must have Country	FAIL	955
GAU-011	MEDIUM	Status consistent with active flag	FAIL	231
GAU-012	MEDIUM	Campaign exists in raw.salesforce_campaign	PASS	0
GAU-013	HIGH	Donation_Item_Code unique	PASS	0
GAU-014	HIGH	Active item has Product_Type	PASS	0
GAU-015	—	HA_Donation_Frequency — distinct values reported (not gated)	 REPORT-ONLY	—
GAU-016	—	Stipulation — distinct values reported (not gated)	 REPORT-ONLY	—
GAU-017	—	Regional_Office_Code — distinct values reported (not gated)	 REPORT-ONLY	—

Rule	Severity	Check	Status	Failed
GAU-018	LOW	Active Sponsorship/Ticket allow-flag set	FAIL	333
GAU-019	MEDIUM	Active item has Status	PASS	0
GAU-020	HIGH	Total_Non_Zakat_Credit not null	PASS	0
GAU-021	HIGH	Total_Zakat_Credit not null	PASS	0
GAU-022	HIGH	funds_available_sadaqa not null	PASS	0
GAU-023	HIGH	funds_available_zakat not null	PASS	0
GAU-024	HIGH	Total_Non_Zakat_Credit numeric	FAIL	8
GAU-025	HIGH	Total_Zakat_Credit numeric	FAIL	2
GAU-026	MEDIUM	npsp__Total_Allocations numeric	FAIL	8

6. Zero-Finding Checks

14 active gated rules returned **0** failures: GAU-001..006, 012..014, 019..023. These confirm identity, uniqueness, financial not-null, and Campaign referential integrity are clean at the current population. The 6 controlled-list rules (GAU-007, 008, 009, 015, 016, 017) are **report-only** — not gated; their distinct values are reported in §7b for stakeholders to confirm (we do not assume any value is wrong).

7. Rule Matches Requiring Review

GAU-010 — Active non-Pledge item missing **Country__c** (HIGH, 955)

955 active, non-Pledge items have no Country. (**Country__c** is 62.6% null overall, but the rule correctly excludes Pledge and inactive items — down from the pre-fix ~17,320.)

SQL check query (top rows):

```
SELECT TOP 10 [Id],[Name],[Product_Type__c],[Country__c],[npsp__Active__c]
FROM staging.item_gau_latest
WHERE LOWER(LTRIM(RTRIM(COALESCE([npsp__Active__c],'false'))))='true'
      AND UPPER(LTRIM(RTRIM(COALESCE([Product_Type__c],''))))<>'PLEDGE'
      AND NULLIF(LTRIM(RTRIM(COALESCE([Country__c],''))),'') IS NULL
ORDER BY [Name];
```

Sample rule matches (actual output — top 8):

Id	Name	Product_Type_c	Country_c	Active
a0Z8e000000GrRzEAK	(DO NOT USE) Water, Sanitation and Hygiene Fund	Fund	NULL	true
a0Z4J000002KDZgUAO	10 Days Food Fund	Fund	NULL	true
a0Z8e000000gPH2EAM	10 Days Food Fund	Fund	NULL	true
a0Z8e000000gPT6EAM	10 Days Food Fund	Fund	NULL	true
a0Z8e000000gUjGEAU	10 Days Food Fund	Fund	NULL	true
a0Z8e000000gV1SEAU	10 Days Food Fund	Fund	NULL	true
a0Z4J000002KDZhUAO	10 Days Orphans and Children Fund	Fund	NULL	true
a0Z8e000000gPH3EAM	10 Days Orphans and Children Fund	Fund	NULL	true

Stakeholder question: Should active non-Pledge items require a Country, or are some global/HQ items exempt?

GAU-011 — Status inconsistent with active flag (MEDIUM, 231)

231 items are `npsp__Active__c = true` while `Status__c = 'Inactive'` — a direct flag/status conflict.

Diagnostic — Status distribution for active items (actual output):

```
status_val | cnt
Active     | 25,602
Inactive   | 231
```

SQL check query (top rows):

```
SELECT TOP 10 [Id],[Name],[npsp__Active__c],[Status__c]
FROM staging.item_gau_latest
WHERE LOWER(LTRIM(RTRIM(COALESCE([npsp__Active__c], 'false'))))='true'
  AND UPPER(LTRIM(RTRIM(COALESCE([Status__c], '')))) IN
('INACTIVE', 'CLOSED', 'ARCHIVED')
ORDER BY [Name];
```

Stakeholder question: Which field is authoritative — `npsp__Active__c` or `Status__c`? (All 231 are active-flagged but status=Inactive.)

GAU-018 — Active Sponsorship/Ticket with no allow-flag (LOW, 333)

333 active Sponsorship/Ticket items have both `Allow_Single__c` and `Allow_Recurring__c` false (scoped to those two types by design — the sample is dominated by ticket items).

SQL check query (top rows):

```
SELECT TOP 10 [Id],[Name],[Product_Type__c],[Allow_Single__c],[Allow_Recurring__c]
FROM staging.item_gau_latest
WHERE LOWER(LTRIM(RTRIM(COALESCE([npsp__Active__c], 'false'))))='true'
      AND UPPER(LTRIM(RTRIM(COALESCE([Product_Type__c], '')))) IN
('SPONSORSHIP', 'TICKET')
      AND LOWER(LTRIM(RTRIM(COALESCE([Allow_Single__c], 'false'))))='false'
      AND LOWER(LTRIM(RTRIM(COALESCE([Allow_Recurring__c], 'false'))))='false'
ORDER BY [Name];
```

Sample rule matches (actual output — top 8):

Id	Name	Product_Type__c	Allow_Single__c	Allow_Recurring__c
a0Z8e000000poiyEAA	Adult Entrance Ticket	Ticket	false	false
a0Z8e000000UcH7EAK	Adult Entrance Ticket	Ticket	false	false
a0Z8e000000UOCtEAO	Adult Entrance Ticket	Ticket	false	false
a0Z8e000000UOCuEAO	Adult Entrance Ticket	Ticket	false	false
a0Z8e000000UOCvEAO	Adult Entrance Ticket	Ticket	false	false
a0Z8e000000UOCyEAO	Adult Entrance Ticket	Ticket	false	false
a0Z8e000000UODDEA4	Adult Entrance Ticket	Ticket	false	false
a0ZN200000000tqvMAA	Black Friday Family (5 Tickets)	Ticket	false	false

Stakeholder question: Should active Sponsorship/Ticket items with both allow-flags false accept donations at all?

18 items carry values that fail a `DECIMAL` cast in `Total_Non_Zakat_Credit__c` (8), `Total_Zakat_Credit__c` (2), `npsp__Total_Allocations__c` (8).

Root cause (engineering finding): these are **not garbage** — they are large numbers stored in **scientific / exponential notation** (e.g. `1.31987012E7` = 13,198,701.2). Confirmed:
`TRY_CONVERT(DECIMAL(18,2), '1.31987012E7')` → `NULL`, but
`TRY_CONVERT(FLOAT, '1.31987012E7')` → `13198701.2`. Fix either at source (store plain decimal) **or** relax the rule to parse via `FLOAT` before `DECIMAL`.

SQL check query (GAU-024 shown; swap the column for 025/026):

```
SELECT TOP 10 [Id],[Name],[Total_Non_Zakat_Credit__c]
FROM staging.item_gau_latest
WHERE NULLIF(LTRIM(RTRIM([Total_Non_Zakat_Credit__c])), '') IS NOT NULL
      AND TRY_CONVERT(DECIMAL(18,2),[Total_Non_Zakat_Credit__c]) IS NULL
ORDER BY [Name];
```

Sample rule matches (actual output):

Rule	Id	Name	Raw value (E-notation)
GAU-024	a0Z8e000000gP5fEAE	Gaza Emergency	1.31987012E7
GAU-024	a0Z8e000000gPNgEAM	Gaza Emergency	1.85423558E7
GAU-024	a0Z8e000000UuBUEA0	Gift Aid Income - HMRC	1.396747239E7
GAU-025	a0ZN2000000taKbMAI	100% Zakat for Gaza Emergency	1.340112775E7
GAU-025	a0Z8e000000gPOAEA2	Zakat Fund	1.443536126E7
GAU-026	a0Z4J000002KDiaUAG	Food Fund	1.059147774E7
GAU-026	a0ZN20000000pDpMAI	HA France Project Fund	5.072538999E7
GAU-026	a0Z4J000002KDzFUAW	Where Most Needed	1.170510097E7

Recommendation: normalize the 18 values to plain decimal at source (or widen the rule to `FLOAT`); no business decision needed — mechanical, engineering-safe.

7b. Controlled-list fields — distinct values for stakeholder review

We do **not** assume any of these values is wrong. The rules that used to gate them (GAU-007, GAU-008, GAU-009, GAU-015, GAU-016, GAU-017) are **report-only** (not gated). Below are the actual distinct values and counts — please confirm which are valid / canonical.

Field	Distinct values (count)
Product_Type__c	Pledge 18,322 · Fund 4,557 · Qty-based 2,977 · Project 2,069 · <NULL/BLANK> 1,763 · Sponsorship 510 · Qty-based (Special Request) 315 · Ticket 188 · Medical 59 · Bundle 42 · Deductions 6 · Other Income 5 · Fees Charged 2
Programme_Category__c	General 10,219 · UnAllocated Funds 9,962 · Food 3,493 · Emergency 1,532 · Medical 1,085 · Orphans and Children 713 · Livelihoods 561 · WASH 508 · Education 416 · Development 410 · OCW 382 · EmergencyResponse 347 · SeasonalRamadan 232 · Shelter 214 · SeasonalQurbani 128 · <NULL/BLANK> 125 · Gifts 113 · SpecialRequest 90 · Integrated Development 76 · Children 67 · SeasonalWinterisation 57 · SeasonalRamadan(ZAF) 56 · SeasonalBacktoSchool 26 · Fees Charged 2 · SeasonalEidGifts 1
Donation_Type__c	<NULL/BLANK> 27,928 · Single 1,726 · Single;Recurring 998 · Recurring 163
HA_Donation_Frequency__c	Monthly 29,021 · <NULL/BLANK> 1,786 · Monthly;Daily 8
Stipulation__c	SD 21,481 · SD;ZK 3,742 · SD;ZK;XX 3,539 · <NULL/BLANK> 1,621 · XX 270 · ZK 87 · SD;XX 75
Regional_Office_Code__c	UK 16,305 · All 3,150 · <NULL/BLANK> 1,963 · US 1,802 · CA 1,700 · FR 1,680 · BE 1,178 · ES 1,033 · IE 937 · AR 909 · TR 132 · WQ 26

```
-- Re-run the distinct list for any controlled-list field (change the column)
SELECT COALESCE(NULLIF(LTRIM(RTRIM([Product_Type__c])), ''), '<NULL/BLANK>') AS
value, COUNT(*) AS cnt
FROM staging.item_gau_latest
GROUP BY COALESCE(NULLIF(LTRIM(RTRIM([Product_Type__c])), ''), '<NULL/BLANK>')
ORDER BY cnt DESC;
```

8. Rule Match Summary (ranked)

Totals: 1,537 open exceptions across **1,506 distinct affected items** (overlap: some items fail more than one rule). Only the 6 active gated rules below produced matches; the other 14 gated rules PASS with 0 and the 6 controlled-list rules are report-only.

By severity (open exceptions)

Severity	Count	% of Total
CRITICAL	0	0.0%
HIGH	965	62.8%
MEDIUM	239	15.6%
LOW	333	21.7%

Severity	Count	% of Total
TOTAL	1,537	100.0%

By rule (ranked)

Rank	Rule	Description	Matches	Severity
1	GAU-010	Active non-Pledge item missing Country	955	HIGH
2	GAU-018	Active Sponsorship/Ticket, no allow-flag	333	LOW
3	GAU-011	Status inconsistent with active flag	231	MEDIUM
4	GAU-024	Total_Non_Zakat_Credit not numeric (E-notation)	8	HIGH
5	GAU-026	npsp__Total_Allocations not numeric (E-notation)	8	MEDIUM
6	GAU-025	Total_Zakat_Credit not numeric (E-notation)	2	HIGH
TOTAL			1,537	

9. Business Actions Required

Rule	Count	Stakeholder question	Decision options	Owner	Deadline
GAU-010	955	Must active non-Pledge items have a Country?	Require Country; exempt global/HQ items; remediate at source	Item/GAU Business Owner	This week
GAU-011	231	Which is authoritative — npsp__Active__c or Status__c ?	Trust active flag; trust status; reconcile at source	Item & Process Owner	This week
GAU-018	333	Should active Sponsorship/Ticket with both allow-flags false accept donations?	Set an allow-flag; deactivate; allow as-is	Fundraising Owner	Next week
GAU-024/025/026	18	(Engineering) Normalize E-notation financials?	Fix at source to decimal; widen rule to FLOAT	Data Engineering	This week

Rule	Count	Stakeholder question	Decision options	Owner	Deadline
GAU-007/008/009/015/016/017	report-only	Which controlled-list values are canonical?	Confirm/curate allowed lists (see §7b)	Item/GAU Business Owner	Next week

10. Most-Violating Records

```
SELECT TOP 5 e.record_id, COUNT(DISTINCT r.check_name) AS rules_failed,
      STRING_AGG(r.check_name, ', ') WITHIN GROUP (ORDER BY r.check_name) AS
failed_rules
FROM dq.dq_exceptions e JOIN dq.dq_rule_catalog r ON r.rule_id=e.rule_id
WHERE r.object_name='Item_GAU' AND e.resolution_status='OPEN'
GROUP BY e.record_id ORDER BY rules_failed DESC;
```

Actual output (top 5):

record_id	rules_failed	failed_rules
a0Z8e000000gPLGEA2	3	GAU-018, GAU-024, GAU-026
a0Z8e000000gP3FEAU	2	GAU-018, GAU-024
a0Z8e000000gP9CEAU	2	GAU-018, GAU-026
a0Z8e000000UuBUEA0	2	GAU-024, GAU-026
a0ZN20000000pCDMAY	2	GAU-024, GAU-026

Worst cases are active Sponsorship items that also carry an E-notation financial value (GAU-018 + GAU-024/026).

11. What's Stored Where

Framework status (from live **dq** tables, read 2026-08-01)

- ✅ Item/GAU is registered in the DQ framework — 26 rules in **dq.dq_rule_catalog** (rule_ids 19–24, 53–72).
- ✅ The framework **has executed** (run 2026-07-30) and written **1,537 open Item_GAU exceptions** to **dq.dq_exceptions** (the official DQ layer): GAU-010 = 955, GAU-018 = 333, GAU-011 = 231, GAU-024 = 8, GAU-026 = 8, GAU-025 = 2; the other 14 gated rules PASS with 0.
- ✅ 6 controlled-list rules (GAU-007/008/009/015/016/017) are **report-only** (not gated).
- ⚠️ **Business review / sign-off is still pending** — exceptions remain **OPEN** until reviewed/resolved.
- ❌ **No corrections written back to Salesforce yet** (writeback not enabled).

Table	Rows	Status
staging.item_gau_latest	30,815	Materialized latest-per-Id (31 columns)
dq.dq_exceptions (Item_GAU, OPEN)	1,537	Official DQ layer — awaiting business review
dq.dq_rule_catalog (Item_GAU)	26	20 gated + 6 report-only

12. Sign-off & Next Meeting

What we accomplished

- ☒ Staged 30,815 items (574 duplicates collapsed)
- ☒ Executed 26 rules (20 gated / 6 report-only)
- ☒ Found 1,537 matches across 1,506 items; 0 CRITICAL
- ☒ Identified 4 business decision areas + 1 engineering fix

Decisions needed

- ☐ GAU-010 Country requirement: _____
- ☐ GAU-011 active vs status authority: _____
- ☐ GAU-018 Sponsorship/Ticket allow-flags: _____
- ☐ GAU-024/025/026 E-notation financials — normalize at source or widen rule: _____
- ☐ Controlled-list canonical values (§7b): _____

Date: _____ **Attendees:** _____

13. Recommendations

- **Stakeholder decisions:** GAU-010 (Country requirement), GAU-011 (active vs status authority), GAU-018 (Sponsorship/Ticket allow-flags) — each needs a policy answer before it becomes a hard gate.
- **Controlled-list fields (report-only):** GAU-007/008/009/015/016/017 are report-only — we do not assume valid values. See §7b distinct-value lists and ask stakeholders which are canonical.